

Leading Kazhdan–Lusztig coefficients with fully commutative lower endpoints in type E^*

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Abstract

We prove that the leading coefficient $\mu(x, w)$ of the Kazhdan–Lusztig polynomial $P_{x,w}$ lies in $\{0, 1\}$ whenever x is fully commutative and W is the Weyl group of type E_6 , E_7 or E_8 . Together with the known results in types A and D , this settles for every finite simply laced Coxeter group the question raised by Green in 2009 and called Green’s 0–1 conjecture by Chmutov. The proof reduces, by Green’s star argument as adapted by Gern, to a classification of the terminal (T -avoiding) elements of E_6 , E_7 and E_8 , obtained by exhaustive enumeration with parabolic pruning: the noncommuting terminals are Gern’s type- D list together with exactly two new elements, of lengths 28 in E_7 and 50 in E_8 , and the coefficient statement then follows from Fan’s monomial cell theory, in Green’s formulation, and parity. We also construct infinite families of full-support terminal reflections in every generalized type E_{4r+1} , $r \geq 3$, and in E_{10} , with $\mu(x, b) = 0$ for every fully commutative x and every member of the family, and an affine E_8 family with exact lengths $27 + 92k$ and the same vanishing. The bound fails in affine type D_4 , where $\mu = 2$ occurs, and it fails without the fully commutative hypothesis: $\mu = 10$ occurs already in E_6 .

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1 Introduction

Let (W, S) be a Coxeter system with length function ℓ and Bruhat order $\leq$, and let $P_{x,w}(q)$ be its ordinary equal-parameter Kazhdan–Lusztig polynomials [25]. Put $\mu(x, w) = [q^{(\ell(w)-\ell(x)-1)/2}]P_{x,w}(q)$ if $x < w$ and $\ell(w) - \ell(x)$ is odd, and $\mu(x, w) = 0$ otherwise. A *cover* is a pair $x < w$ with $\ell(w) - \ell(x) = 1$; then $P_{x,w} = 1$ and $\mu(x, w) = 1$. Write $L(w) = \{s \in S : \ell(sw) < \ell(w)\}$ and $R(w) = \{s \in S : \ell(ws) < \ell(w)\}$ for the descent sets and $\text{supp}(w)$ for the set of generators in a reduced word of w ; w has *full support* if $\text{supp}(w) = S$. Following Stembridge [33], w is *fully commutative* (FC) if any two of its reduced words are related by interchanging adjacent commuting generators; in a simply laced system this holds if and only if no reduced word of w contains a factor sts with $m(s, t) = 3$ [33, Proposition 2.1]. The FC elements form the set W_{FC} . A set $I \subseteq S$ of pairwise commuting generators is an *independent set* of the Coxeter graph; $i(I) = \prod_{s \in I} s$ is a *commuting product*. An element is *terminal* if no reduced word for it begins or ends with two noncommuting generators, and a *noncommuting terminal* if moreover it is not a commuting product; these are Gern’s weakly bad and bad elements [16, Definition 2.1.1] and Laird’s T -avoiding and T_2 -avoiding elements [26].

Theorem 1.1 (Finite exceptional types). *If W has type E_6 , E_7 or E_8 , then $\mu(x, w) \in \{0, 1\}$ for all $x \in W_{\text{FC}}$ and $w \in W$.*

Corollary 1.2 (Finite simply laced groups). *If W is a finite simply laced Coxeter group, $x \in W_{\text{FC}}$ and $w \in W$, then $\mu(x, w) \in \{0, 1\}$.*

Our second theme is arbitrarily large rank. Throughout, the generalized diagram E_n , $n \geq 6$, is the chain $0-1-\dots-(n-2)$ with node $n-1$ attached to node 2; thus $E_9 = \widetilde{E}_8$, and E_n is indefinite for $n \geq 10$. Section 4.1 defines, for every $r \geq 3$, explicit real roots $\beta_{r,k}$ ($k \geq 0$) of E_{4r+1} , with reflections $b_{r,k} = r_{\beta_{r,k}}$, and an independent set I_r of size $2r+1$.

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Theorem 1.3 (A rank-uniform family). *For every $r \geq 3$ the reflections $b_{r,k}$, $k \geq 0$, are pairwise distinct full-support terminal elements of $W(E_{4r+1})$ that are not fully commutative, with $L(b_{r,k}) = R(b_{r,k}) = I_r$. For every fully commutative x and every $k \geq 0$, $\mu(x, b_{r,k}) = 0$.*

Theorem 4.5 gives an analogous family in $\tilde{E}_8$ with exact lengths $27 + 92k$, and Proposition 4.6 one in E_{10} ; so the terminal obstructions to length-decreasing star reduction cannot be exhausted by a finite list in E_n for $n \geq 9$.

Green [21, Introduction] asked whether $\mu(x, w) \in \{0, 1\}$ for fully commutative x in a large class of Coxeter groups, “possibly all of them”; Chmutov [7, abstract and §2.3] calls this *Green’s 0–1 conjecture*. It is known in types A and $\tilde{A}$ (Green [21, Theorem 4.1; Theorem 3.1 of the arXiv version]), $B = C$ (Chmutov [7, Theorem 2.3.8]) and D (Gern [16, Theorem 4.5.11]); Theorem 1.1 completes the finite simply laced case, F_4 , H_3 and H_4 remain open (dihedral groups, G_2 included, have all $P_{x,w} = 1$), and finiteness cannot be dropped from Corollary 1.2, since $\mu = 2$ occurs in $\tilde{D}_4$ (Section 5).

Jones [24, Theorems 1.2 and 6.1] proved the bound for a Deodhar upper endpoint in every finite Weyl group (E_6 , E_7 , E_8 , F_4 and G_2 by computer), the only earlier exceptional 0–1 result, with the opposite endpoint hypothesis; when both endpoints are fully commutative the bound holds in every E_n by Graham’s thesis, as recalled in [20, Remark 9.5] and [19, Remark 7.12]. The reduction to terminal upper endpoints (Lemma 2.2) is Green’s argument as adapted by Gern, and the lower-endpoint lemma (Lemma 2.3) is Fan’s monomial cell theorem [12] in Green’s formulation [20]; what is new is the classification of the terminal elements of E_6 , E_7 and E_8 (Proposition 3.1) by exhaustive enumeration, with parabolic pruning in place of the 696,729,600 elements of E_8 . The noncommuting terminals turn out to be Gern’s type- D list together with exactly two new elements, $w_7 \in E_7$ of length 28 and $w_8 \in E_8$ of length 50 (Remark 3.3); each has a unique fully commutative lower endpoint satisfying both descent inclusions, with even length gap for the new ones, so the only Kazhdan–Lusztig value needed is Gern’s D_6 value $\mu(x_6, w_6) = 1$ [16, Lemma 4.5.5] (du Cloux’s Coxeter; recomputed in Appendix A).

Infinite families of T -avoiding elements inside a fixed infinite Coxeter group were known in affine type A (Fan–Green [13, Proposition 3.1.2], fully commutative), affine type C (Ernst [10]; Laird [26, Theorem 4.1.5]) and type F_n (non-FC; Cross, Ernst, Hills–Kimball and Quaranta (2012) and Gilbertson–Ernst (2013), as reported in [26, §3.3]); Laird [26, §5.3] lists the affine type- E questions as open. By Green’s theorem [18, Theorem 6.3] every E_n is star reducible, so every fully commutative terminal in E_n is a commuting product, which is why all families here are non-FC.

Exceptional Kazhdan–Lusztig computations feature in Geck’s *PyCox* [14] and in Geck–Halls’s analysis of the cells of E_8 [15], which concern cell partitions and character values rather than every $\mu(x, w)$ with FC lower endpoint; fully commutative cell closure fails in type E (Green–Losonczy [22, Example 2.2.5], credited to Graham [17, §9.9]), and a W -graph records $\mu(x, w)$ only through a generator in $L(x) \setminus L(w)$ [15, Remark 2.1], whereas our residual terminal pairs have equal descent sets. To our knowledge, the exceptional bound, the terminal lists of Table 1 and the infinite families have not appeared before.

Sections 2, 3 and 4 contain the structural lemmas, the finite theorem and the infinite families, Section 5 complements and open problems, and Appendix A the computational certificates.

2 Structural reductions

2.1 Star reduction

We use the elementary properties of μ from [25, §2.3 and Theorem 4.2], restated in [16, Proposition 1.2.4].

Lemma 2.1 (Descent criterion). *Let $x < w$. If $s \in L(w) \setminus L(x)$, then $\mu(x, w) = 0$ unless $x = sw$; if $s \in R(w) \setminus R(x)$, then $\mu(x, w) = 0$ unless $x = ws$.*

Proof. This is [25, (2.3.e) and (2.3.f)]; see [16, Proposition 1.2.4(3),(4)]. □

For noncommuting generators s, t with $m(s, t) = 3$ let $D_L(s, t) = \{w : |L(w) \cap \{s, t\}| = 1\}$. For $w \in D_L(s, t)$ exactly one of sw, tw lies in $D_L(s, t)$; it is the left star $*w$. Right stars w^* are defined symmetrically. Stars

preserve full commutativity in simply laced systems [32, Proposition 2.10], [16, Proposition 1.1.29]. Star transport [25, Theorem 4.2] says that for $x, w \in D_L(s, t)$,

$$\tilde{\mu}(x, w) = \tilde{\mu}(*x, *w), \quad (1)$$

where $\tilde{\mu}(y, z)$ equals $\mu(y, z)$ if $y < z$, $\mu(z, y)$ if $z < y$, and 0 otherwise; the same holds for right stars.

Lemma 2.2 (Terminal reduction). *Let W be simply laced. If $\mu(x, b) \in \{0, 1\}$ for every terminal upper endpoint b and every fully commutative x , then $\mu(x, w) \in \{0, 1\}$ for all $x \in W_{\text{FC}}$ and all $w \in W$.*

Proof. This is the argument of Green [21, proof of Theorem 3.1 (published Theorem 4.1)], as adapted by Gern [16, Lemma 2.3.10 and Theorem 4.5.11]. We induct on $\ell(w)$. Put $d = \ell(w) - \ell(x)$; only $x < w$ with $d \geq 3$ odd needs proof. If $L(w) \not\subseteq L(x)$ or $R(w) \not\subseteq R(x)$, Lemma 2.1 gives zero; so assume both inclusions, whence $L(w)$ and $R(w)$ are commuting, like the descent sets of the fully commutative x . If w is terminal, the hypothesis applies. Otherwise let $w = stv$ be reduced with $m(s, t) = 3$ (a noncommuting suffix is treated with right stars). Then $s \in L(w)$, $t \notin L(w)$ because $L(w)$ is commuting, and $s, t \notin L(v)$ (t by reducedness, and $s \in L(v)$ would give $w = stsv' = tsv'$ and $t \in L(w)$); so $\{tv, stv\}$ is the left $\{s, t\}$ -string in $D_L(s, t)$ and $*w = sw$ has length $\ell(w) - 1$. Descent inclusion gives $s \in L(x)$ and full commutativity $t \notin L(x)$, so $x \in D_L(s, t)$ and $*x$ is defined and fully commutative. By (1), $\mu(x, w) = \tilde{\mu}(*x, *w)$; as $\ell(*x) = \ell(x) \pm 1$, the new length difference is d or $d - 2$, both positive, so the value is zero if the new endpoints are incomparable and lies in $\{0, 1\}$ by induction otherwise. $\square$

2.2 Maximum commuting descents

The monomial cell structure of the Temperley–Lieb quotient in type E_n is due to Fan [12]; we use Green’s formulation [20, §4] (statement numbers follow arXiv:0704.0283v1). The next lemma makes explicit the singleton-cell argument ending Green’s proof of [20, Lemma 5.2]; its type- D case is [16, Lemma 4.5.8].

Lemma 2.3 (Maximum common commuting descents). *Let $n \geq 6$, $W = W(E_n)$, and let $x \in W_{\text{FC}}$ with $\text{supp}(x) \subseteq K$. Let I be an independent set of maximum cardinality in the subgraph induced by K . If $I \subseteq L(x) \cap R(x)$, then $x = i(I)$.*

Proof. Let $a_{\text{TL}}(x)$ be the maximum size of a commuting product occurring as a length-additive factor of x [20, Definition 4.1], Fan’s a -function [12, Definition 2.3.1, p. 141]. The descent inclusions give a prefix and a suffix $i(I)$, so $a_{\text{TL}}(x) \geq |I|$, while support and maximality give $a_{\text{TL}}(x) \leq |I|$. By [20, Proposition 4.4(iv)], which is [12, Proposition 4.4.3, p. 148], and its left-hand version (Fan’s dual statement, p. 148), x lies in both the monomial right cell and the monomial left cell of $i(I)$, and by [20, Proposition 4.4(v)], that is [12, p. 140 and proof of Theorem 6.1.2, p. 154], this intersection is $\{i(I)\}$. Fan proves these statements for the whole series E_n , $n \geq 6$; no identification of monomial cells with Kazhdan–Lusztig cells is used. $\square$

Proposition 2.4 (Maximum-descent parity criterion). *Let $n \geq 6$ and let $b \in W(E_n)$ satisfy $L(b) = R(b) = I$, where I is a maximum independent set in the subgraph induced by $\text{supp}(b)$. If $\ell(b) \equiv |I| \pmod{2}$, then for every $x \in W_{\text{FC}}$,*

$$\mu(x, b) \in \{0, 1\}, \quad x < b, \ell(b) - \ell(x) > 1 \implies \mu(x, b) = 0.$$

Proof. The type- D prototype is [16, Lemmas 4.5.4 and 4.5.9]. Let $x < b$ be fully commutative with $\ell(b) - \ell(x) > 1$ and $\mu(x, b) \neq 0$. By Lemma 2.1, $I \subseteq L(x) \cap R(x)$; the subword property gives $\text{supp}(x) \subseteq \text{supp}(b)$, so Lemma 2.3 with $K = \text{supp}(b)$ forces $x = i(I)$. Then $\ell(b) - \ell(x) = \ell(b) - |I|$ is even and $\mu(x, b) = 0$, a contradiction. Covers have $\mu = 1$. $\square$

Remark 2.5. Proposition 2.4 does not require b to be terminal. The parity hypothesis cannot be dropped: Section 5.3 exhibits elements of $\tilde{E}_8$ satisfying every other hypothesis with $\mu(i(I), b) = 1$.

Remark 2.6. The independence number of E_n is $\lceil n/2 \rceil$: excluding node 2 leaves three paths with maximum total $2 + \lceil (n-4)/2 \rceil$, and including it gives $2 + \lceil (n-5)/2 \rceil$, the larger of the two being $\lceil n/2 \rceil$.

2.3 Geometric tests and covers of terminal reflections

Normalize simple roots by $(\alpha_j, \alpha_j) = 2$ and $(\alpha_s, \alpha_t) = -1$ for adjacent nodes, and write $s \sim t$ when s and t are adjacent, i.e. $m(s, t) = 3$. For a real root β the reflection r_β acts by $r_\beta(\rho) = \rho - (\rho, \beta)\beta$; W acts on the left and the matrix of w has columns $w(\alpha_j)$. Root inequalities refer to the simple-root basis, and $s \in R(w)$ if and only if $w(\alpha_s) < 0$. Since $(ws)(\alpha_t) = w(\alpha_s + \alpha_t)$ for $s \sim t$, the right-terminal test is

$$s \in R(w) \implies w(\alpha_s + \alpha_t) > 0 \quad \text{for every } t \sim s, \quad (2)$$

and testing w^{-1} gives the left-terminal condition. The *depth* $\text{dp}(\beta)$ of a positive root β is the least d such that $\beta = s_1 \cdots s_{d-1} \alpha_t$ for simple reflections s_i and a simple root α_t (cf. [3, §4.6]).

Lemma 2.7 (Reflections). *Let (W, S) be simply laced, let β be a positive root and $d = \text{dp}(\beta)$.*

- (i) $\ell(r_\beta) = 2d - 1$, and for every expression $\beta = s_1 \cdots s_{d-1} \alpha_t$ of minimal length the palindromic word $s_1 \cdots s_{d-1} t s_{d-1} \cdots s_1$ is a reduced word for r_β .
- (ii) If $s \in S$ and $(\beta, \alpha_s) < 0$, then $s\beta$ is a positive root of depth $d+1$, $\ell(sr_\beta) = \ell(r_\beta) + 1$ and $\ell(sr_\beta s) = \ell(r_\beta) + 2$. If $(\beta, \alpha_s) > 0$ and $\beta \neq \alpha_s$, then $s\beta$ is a positive root of depth $d - 1$.
- (iii) r_β is fully commutative if and only if β is simple.

Proof. Recall that $\ell(ws) > \ell(w)$ if and only if $w(\alpha_s) > 0$, and that s negates only $\pm\alpha_s$. Let $(\beta, \alpha_s) = -c < 0$. Then $s\beta = \beta + c\alpha_s > 0$; $r_\beta(\alpha_s) = \alpha_s + c\beta > 0$ gives $s \notin R(r_\beta) = L(r_\beta)$, so $\ell(sr_\beta) = \ell(r_\beta) + 1$; and $(sr_\beta)(\alpha_s) = -\alpha_s + cs\beta = c\beta + (c^2 - 1)\alpha_s > 0$ gives $\ell(sr_\beta s) = \ell(r_\beta) + 2$. As $sr_\beta s = r_{s\beta}$, this shows $\ell(r_{s\beta}) = \ell(r_\beta) + 2$ when $(\beta, \alpha_s) < 0$ and, applied to $s\beta$, $\ell(r_{s\beta}) = \ell(r_\beta) - 2$ when $(\beta, \alpha_s) > 0$, $\beta \neq \alpha_s$. Now $\ell(r_\beta) = 2\text{dp}(\beta) - 1$ by induction on the odd number $\ell(r_\beta)$ (trivial for $\ell(r_\beta) = 1$): if $\ell(r_\beta) \geq 3$, some s has $(\beta, \alpha_s) > 0$, $\beta \neq \alpha_s$ (as $(\beta, \beta) = 2$, $\beta \geq 0$), so $\beta' = s\beta > 0$ has $\ell(r_{\beta'}) = \ell(r_\beta) - 2 = 2\text{dp}(\beta') - 1$ and $\text{dp}(\beta) \leq \text{dp}(\beta') + 1$ gives $2\text{dp}(\beta) - 1 \leq \ell(r_\beta)$, while the palindromic word gives $\ell(r_\beta) \leq 2d - 1$. This proves (i); (ii) follows from (i) and the two length formulas. For (iii), if $d \geq 2$ the reduced word of (i) contains $s_{d-1} t s_{d-1}$, which equals t unless $m(s_{d-1}, t) = 3$; so it is a braid factor and $r_\beta \notin W_{\text{FC}}$ [33, Proposition 2.1]. $\square$

Lemma 2.8 (Covers of terminal reflections). *Let $n \geq 6$, $W = W(E_n)$, and let $b = r_\beta$ for a positive root β with at least three nonzero coordinates. Suppose $L(b) = R(b) = I$, where I is a maximum independent set of the subgraph induced by $\text{supp}(b)$ and $|I|$ is odd. Then no Bruhat cover of b is fully commutative. Consequently $\mu(x, b) = 0$ for every fully commutative x .*

Proof. Let $x < b$ with $\ell(x) = \ell(b) - 1$. If $I \subseteq L(x) \cap R(x)$ and $x \in W_{\text{FC}}$, then $\text{supp}(x) \subseteq \text{supp}(b)$ and Lemma 2.3 gives $x = i(I)$ of odd length $|I|$, impossible as $\ell(b)$ is odd. Otherwise some $s \in I$ has $s \notin L(x)$ (or, symmetrically, $s \notin R(x)$); the lifting property [3, Proposition 2.2.7] gives $x \leq sb$, hence $x = sb = (bs)^{-1}$ by length and involutivity, and it suffices to show $bs \notin W_{\text{FC}}$. Put $m = (\alpha_s, \beta)$; $s \in R(b)$ gives $b(\alpha_s) = \alpha_s - m\beta < 0$, so $m \geq 1$, and $s\beta = \beta - m\alpha_s$ is a positive non-simple root, its coordinates off s being those of β . Since $sbs = r_{s\beta}$ and $(\alpha_s, s\beta) = -m < 0$, Lemma 2.7(ii) gives $\ell(sr_{s\beta}) = \ell(r_{s\beta}) + 1$, so s followed by the palindromic word of Lemma 2.7(i) for $r_{s\beta}$ is a reduced word for bs containing a braid factor; hence $bs \notin W_{\text{FC}}$. Non-covers are handled by Proposition 2.4, since $\ell(b)$ and $|I|$ are odd. $\square$

3 Finite exceptional types

3.1 The terminal classification

A fully commutative x is *eligible* for a terminal b if $x \leq b$, $L(b) \subseteq L(x)$ and $R(b) \subseteq R(x)$; by Lemma 2.1, every ineligible non-cover has $\mu(x, b) = 0$. Elements are written as strings of node labels in the numbering of Section 1: the chain $0-1-\cdots-(n-2)$ with node $n-1$ attached to node 2.¹ A string denotes the product of the corresponding simple reflections in the order written, with the left action and column convention of Section 2.3. Every element displayed in Table 1 and in the affine family of Section 4.2 is an involution, so a string and its reverse denote the same element.

¹In Bourbaki's labelling of E_n (chain $1-3-4-\cdots-n$, node 2 attached to node 4), our node 0 is Bourbaki's node 1, our node j with $1 \leq j \leq n-2$ is Bourbaki's node $j+2$, and our node $n-1$ is Bourbaki's node 2.

Proposition 3.1 (Computational classification). *The noncommuting terminals in E_6 , E_7 and E_8 are exactly the eleven elements in Table 1. Both descent sets of each of them equal the support of the commuting product x shown in the table. The remaining terminals are commuting products: 22, 36 and 58 of them, respectively, including the identity.*

Extending Gern's notation [16, Theorem 2.3.6], we write w_4 and w_6 for his elements of lengths 7 and 15 (viewed inside E_n as in Remark 3.3), and w_7 , w_8 for the two new elements of lengths 28 and 50, where w_8 is the concatenation of the two strings

$$(7534231270123456210321432)(5437210321432543721324357).$$

Type	Terminal b	$\ell(b)$	$x = i(I)$	$\ell(b) - \ell(x)$	Source
E_6	1325213	7	135	4	w_4
E_7	1326213	7	136	4	w_4
E_7	13256213	8	1356	4	$w_4 s_5$
E_7	132543621324356	15	1356	11	w_6
E_7	1325436210321432543621324356	28	1356	24	w_7
E_8	1327213	7	137	4	w_4
E_8	13257213	8	1357	4	$w_4 s_5$
E_8	61327213	8	1367	4	$w_4 s_6$
E_8	132543721324357	15	1357	11	w_6
E_8	1325437210321432543721324357	28	1357	24	w_7
E_8	w_8 (displayed above)	50	1357	46	w_8

Table 1: The noncommuting terminals of E_6 , E_7 , E_8 : a reduced word, the length, the unique eligible fully commutative lower endpoint (Corollary 3.2), the gap, and the identification of Remark 3.3. Only the rows of length 15, Gern's D_6 pair, have odd gap.

Corollary 3.2. *Each terminal in Table 1 has the unique eligible fully commutative lower endpoint x displayed there.*

Proof of Corollary 3.2. The induced support graphs have types D_4 , $D_4 \sqcup A_1$, D_6 , E_7 and E_8 , with independence numbers 3, 4, 4, 4, 4: the matchings (1, 2) in the rows of lengths 7 and 8, (1, 2), (3, 4) in the rows of length 15, (0, 1), (2, 3), (4, 5) in the rows of length 28 and (0, 1), (3, 4), (5, 6), (2, 7) in the row of length 50 bound every independent set by the number of support nodes minus the matching size, and the displayed descent set attains the bound (compare Remark 2.6). If y is eligible, then $\text{supp}(y) \subseteq \text{supp}(b)$ and Lemma 2.3 forces $y = x$; conversely x , a commuting product of left descents of b , is a reduced prefix of b , hence below b and eligible. $\square$

Remark 3.3 (Gern's list plus two). The parabolic subgroup of E_n on $\{1, n-1, 2, \dots, n-2\}$ has type D_{n-1} ; sending Gern's labels $1, 2, j$ ($3 \leq j \leq 7$) to $1, n-1, j-1$ identifies it with Gern's $W(D_{n-1})$, and then the rows of Table 1 of lengths 7, 8 and 15 are Gern's bad elements w_4 , $w_4 u$ and w_6 of [16, Theorem 2.3.6], while the E_8 row of length 28 is w_7 in the parabolic subgroup on $\{0, \dots, 5, 7\}$. The E_n form of [16, Theorem 2.3.6] holds for $n = 6, 7, 8$: the noncommuting terminals of E_n are exactly the $w_k u$ with $k \in \{4, 6, 7, 8\}$ and $w_k \in W(E_n)$ (so $k = 4$ for $n = 6$ and $k \leq n$ for $n = 7, 8$), and u a commuting product of generators neither in nor adjacent to $\text{supp}(w_k)$; for $n = 7, 8$, w_n is the unique longest one (matrix equality; exhaustive enumeration of D_5 , D_6 , D_7 for the type- D parabolics). Each w_k is an involution with $L = R = I$: w_4 a reflection, w_6 a product of three orthogonal reflections, w_7 and w_8 of involution types $4A_1$ and $2A_1$; the palindromic layered words, the negated roots and the right weak order chain $w_4 < w_6 < w_7 < w_8$ are in the certificate (Appendix A). For $b = w_7, w_8$, $\mu(i(I), b) = 0$ also follows without parity: $J = I \cup \{0\}$ has type $A_2 \times A_1^3$ and $\ell(w_0(J)) = 6$, and $b = u w_0(J) v$ length-additively with lengths $11 + 6 + 11$ and $11 + 6 + 33$ (words in the certificate), so $b \leq_{LR} w_0(J)$ [25, §2.3], $a(b) \geq a(w_0(J)) = 6$ for Lusztig's a -function [28, Proposition 13.8 and Conjecture 14.2 (P4), (P12), proved for finite Weyl groups in §15.1], [16, Lemmas 1.3.7 and 1.3.8], and, as $I \subseteq L(b)$, $P_{i(I), b} = P_{e, b}$ [25, §2.3] has degree at most $(\ell(b) - a(b))/2 \leq (\ell(b) - 6)/2$ [16, Proposition 1.3.5], citing [27, 1.3(a)], below the degree $(\ell(b) - 5)/2$ of $\mu(i(I), b)$. Likewise $a(w_6) \geq 5$, consistent with $a(w_6) = 5$ [16, Proposition 4.5.1].

3.2 Exhaustiveness, and the proofs of Theorem 1.1 and Corollary 1.2

All computations use the faithful integral geometric representation: right multiplication by s_i replaces the columns $a_j = w(\alpha_j)$ by $a_i \mapsto -a_i$ and $a_j \mapsto a_j + a_i$ for $j \sim i$; signs give descents and (2) tests terminality on both sides. In E_6 and E_7 , breadth-first closure under all generators enumerates the whole group. For E_8 , fix $J \subseteq S$ and write $w = av$ length-additively with $a \in W^J$ (minimal coset representatives) and $v \in W_J$; call w *right-terminal* if no reduced word of w ends in a noncommuting pair.

Lemma 3.4 (Parabolic pruning). *If $w = av$ is right-terminal, then v is right-terminal in W_J .*

Proof. A reduced word for v ending in adjacent generators, prefixed by a reduced word for a , gives such a word for w , by length additivity. $\square$

So one enumerates the products av with v right-terminal, applies (2) in the ambient group and tests inverses. The implementation recurses along $E_8 \supset E_7 \supset E_6 \supset D_5 \supset D_4 \supset A_3 \supset A_2 \supset A_1 \supset \{e\}$, obtaining the minimal coset representatives at each step by closure under left multiplication and stripping of right J -descents (W acts transitively on W/W_J); Lemma 3.4 and induction give completeness at every stage. The final E_7 parabolic has 576 right-terminals and 240 cosets, their products give 2,160 right-terminals in E_8 , and the inverse test leaves exactly the 64 terminals of Proposition 3.1. With Corollary 3.2, the finite theorem thus needs neither an enumeration of pairs with FC lower endpoint nor a table of Kazhdan–Lusztig polynomials.

Proof of Theorem 1.1. By Lemma 2.2 it suffices to treat terminal b . If b is a commuting product, every $x < b$ is a proper subproduct, so some $s \in L(b) \setminus L(x)$ exists and Lemma 2.1 gives $\mu(x, b) \in \{0, 1\}$. For a noncommuting terminal, Corollary 3.2 and Lemma 2.1 leave one candidate non-cover pair (x, b) per row of Table 1, and parity disposes of all rows except the two of length 15. These are the same pair in a D_6 parabolic subgroup: in E_n , $n = 7, 8$, mapping Gern’s labels $1, \dots, 6$ to $1, n-1, 2, 3, 4, 5$ identifies it with Gern’s $x_6 = (-1, -2, 4, 3, 6, 5)$ and $w_6 = (-1, -6, 3, -4, 5, -2)$ in signed permutation notation [16, Example 1.1.6]. Bruhat order and Kazhdan–Lusztig polynomials of a standard parabolic subgroup agree with those of W , since reduced words of W_J are reduced in W and the Hecke algebra of W_J is a bar-stable subalgebra containing its Kazhdan–Lusztig basis [3, Chapter 5]. Gern’s Lemma 4.5.5 [16], a computation with du Cloux’s Coxeter, gives $\mu(x_6, w_6) = 1$ (recomputed in Appendix A). Hence every coefficient with terminal upper endpoint lies in $\{0, 1\}$. $\square$

Proof of Corollary 1.2. The irreducible finite simply laced types are A , D and E ; combine Green’s theorem, Gern’s theorem and Theorem 1.1. For a direct product $W = \prod_j W_j$, full commutativity is componentwise and $P_{x,w} = \prod_j P_{x_j, w_j}$. Let $x < w$, let d_j be the component length differences, $d = \sum_j d_j$, and m the number of j with $d_j > 0$. Even d gives $\mu = 0$; for odd d and $m \geq 2$, $\deg P_{x_j, w_j} \leq (d_j - 1)/2$ gives $\deg P_{x,w} \leq (d - m)/2 < (d - 1)/2$, so $\mu = 0$; for $m = 1$, $\mu(x, w)$ is the coefficient in the single nontrivial component, which lies in $\{0, 1\}$. $\square$

4 Infinite families

4.1 The rank-uniform construction

Fix an integer $r \geq 3$, put $n = 4r + 1$, and use the E_n numbering of Section 1, so node $4r$ is attached to node 2. Define β_r by its coordinates $(\beta_r)_0 = r - 1$, $(\beta_r)_1 = r$, $(\beta_r)_2 = 2r - 1$, $(\beta_r)_{4r} = r$ and $(\beta_r)_j = 2r - \lfloor j/2 \rfloor$ for $3 \leq j \leq 4r - 1$, where $(v)_j$ denotes the coefficient of α_j in v ; so the long-arm coordinates are $2r - 1, 2r - 2, 2r - 2, \dots, 1, 1$. Put $I_r = \{0, 3, 5, \dots, 4r - 1, 4r\}$ and $a = 2r - 4$. On the embedded affine E_8 subdiagram $\{0, \dots, 7, 4r\}$ set $(\delta_0, \dots, \delta_7) = (2, 4, 6, 5, 4, 3, 2, 1)$, $\delta_{4r} = 3$, $(\gamma_0, \dots, \gamma_7) = (1, 2, 3, 2, 2, 1, 1, 0)$, $\gamma_{4r} = 1$, with all other coordinates zero. Set $T = r_\gamma r_{\gamma+\delta}$ and, for $k \geq 0$,

$$\beta_{r,k} = \beta_r - ak\gamma + (ak^2 + rk)\delta, \quad b_{r,k} = r_{\beta_{r,k}}. \quad (3)$$

Lemma 4.1. *For every $r \geq 3$, β_r is a real root of E_{4r+1} .*

Proof. Deleting node 0 leaves finite type D_{4r} ; realize its simple roots in an orthonormal basis $e_1, \dots, e_{4r}$ as $\alpha_1 = e_1 + e_2$, $\alpha_{4r} = e_2 - e_1$, $\alpha_j = e_{j+1} - e_j$ ($2 \leq j \leq 4r-1$), take z orthogonal to them with $(z, z) = 2 - r$, and put $\alpha_0 = z - \frac{1}{2} \sum_i e_i$; these vectors have the Cartan inner products of E_{4r+1} and form a basis, the Cartan determinant $9 - n$ being nonzero. Then $\beta_r = (r-1)z + \sum_i c_i e_i$ with $c_i = -(r-1)/2$ for odd and $(3-r)/2$ for even i , and $(\beta_r, \alpha_0) = r-2$ gives $s_0 \beta_r = F \alpha_0$, where F flips the $2r$ even coordinates. As $F = \prod_{j=1}^r r_{e_{4j-2}-e_{4j}} r_{e_{4j-2}+e_{4j}} \in W(D_{4r})$ [16, Example 1.1.6], $\beta_r = s_0 F \alpha_0$ is a real root. $\square$

Lemma 4.2. *Suppose γ and $\gamma + \delta$ are real roots with $(\gamma, \gamma) = 2$ and $(\gamma, \delta) = (\delta, \delta) = 0$, and let $T = r_\gamma r_{\gamma+\delta}$. Then*

$$(T-1)\rho = (\rho, \delta)\gamma - ((\rho, \gamma) + (\rho, \delta))\delta, \quad (T-1)^3 = 0,$$

and if $(\beta, \delta) = -a$ and $(\beta, \gamma) = -c$, then

$$T^k \beta = \beta - ak\gamma + (ak^2 + ck)\delta \quad (k \geq 0). \quad (4)$$

Proof. Compose the two reflection formulas. With $N = T - 1$, $N\delta = 0$ and $N\gamma = -2\delta$ give $N^3 = 0$, and $N\beta = -a\gamma + (a+c)\delta$, $N^2\beta = 2a\delta$; expand $(1+N)^k$ binomially. The lemma holds on the whole ambient root space, where δ need not be in the radical. $\square$

Lemma 4.3. *For all $r \geq 3$ and $k \geq 0$, $\beta_{r,k} = T^k \beta_r$ is a real root with every coordinate strictly positive. Its reflection $b_{r,k}$ is terminal and has both descent sets equal to I_r .*

Proof. The vector γ is a root of the finite E_8 parabolic subsystem obtained by omitting node 7, and $\gamma + \delta$ is an affine real root (lowering words to a simple root are recorded in the certificate, Appendix A). As $(\beta_r, \gamma) = -r$ and $(\beta_r, \delta) = -a$, (4) gives $T^k \beta_r = \beta_{r,k}$, a real root, and the binomial increments $(a+r)\delta - a\gamma$ and $2a\delta$ are coordinatewise nonnegative for $r \geq 3$, so every coordinate of $\beta_{r,k}$ is strictly positive. The pairings $m_j = (\alpha_j, \beta_{r,k})$ are

$$\begin{aligned} m_0 &= r-2, & m_1 &= 2-r, & m_2 &= -1-ak, & m_8 &= -1-ak^2-rk, & m_{4r} &= 1+ak, \\ m_j &= (-1)^{j+1}(1+ak) \quad (3 \leq j \leq 7), & m_j &= (-1)^{j+1} \quad (9 \leq j \leq 4r-1), \end{aligned}$$

positive exactly on I_r . Since $\beta_{r,k} > 0$, $b_{r,k}(\alpha_j) = \alpha_j - m_j \beta_{r,k}$ is negative exactly when $m_j > 0$, so $R(b_{r,k}) = I_r = L(b_{r,k})$ by involutivity. For every edge $s \sim t$ with $s \in I_r$ the sum $m_s + m_t$ is nonpositive (at $7-8$ it is $-ak(k-1)-rk$, at $8-9$ it is $-ak^2-rk$, elsewhere by inspection), so $b_{r,k}(\alpha_s + \alpha_t) = \alpha_s + \alpha_t - (m_s + m_t)\beta_{r,k} > 0$, which is (2); the left test follows by involutivity. $\square$

Proof of Theorem 1.3. The set I_r is independent of size $2r+1$, and the matching $(0, 1), (2, 4r), (3, 4), (5, 6), \dots, (4r-3, 4r-2)$ with $2r$ edges leaves one vertex unmatched, so I_r is a maximum independent set of E_{4r+1} (compare Remark 2.6). By Lemma 4.3, $b_{r,k}$ is a terminal reflection with $L = R = I_r$, $|I_r|$ is odd, and $\beta_{r,k}$ has $4r+1 \geq 3$ nonzero coordinates, so Lemma 2.8 gives $\mu(x, b_{r,k}) = 0$ for every fully commutative x , and $b_{r,k} \notin W_{\text{FC}}$ by Lemma 2.7(iii). Since $m_0 = r-2 > 0$ and $\beta_{r,k}$ is positive in every coordinate, column 0 of $b_{r,k}$ changes every coordinate row, whereas omitting a generator would leave its row unchanged; so $b_{r,k}$ has full support. Distinct positive roots give distinct reflections, and $(\beta_{r,k})_0 = r-1+4k+(4r-8)k^2$ gives distinctness. $\square$

Remark 4.4. At $r=3$ the seed is $\beta_3 = (2, 3, 5, 5, 4, 4, 3, 3, 2, 2, 1, 1, 3)$ and $\ell(b_{3,0}) = 75$, while the maximum lengths of fully commutative elements of $E_{10}, \dots, E_{13}$ are 55, 66, 78 and 92 (Appendix A), so a length-based exclusion of fully commutative covers already fails in rank 13; the lengths $\ell(b_{r,k}) = 8r^2 + 3 + 116k$ hold for every pair in the certificate ($r \leq 8, k \leq 3$; the supplement's optional extended run adds $r = 9, 10, 12$) but are not needed. The restriction $r \geq 3$ is structural: at $r=2$, $\beta_2 = \delta - \gamma$, $(\alpha_0, \beta_2) = 0$, and the descent set has even size 4.

4.2 The affine family

Use E_9 numbering: chain $0-1-\dots-7$ with node 8 attached to 2; the affine node is 7. Put

$$\delta' = (2, 4, 6, 5, 4, 3, 2, 1, 3), \quad \gamma' = (1, 1, 1, 1, 1, 0, 0, 0, 0), \quad I' = \{1, 3, 5, 7, 8\},$$

and

$$c_0 = 312875645234123012856745231,$$

a reduced word for an involution. Let $T' = r_{\gamma'} r_{\gamma' + \delta'}$, so that $T'(\alpha) = \alpha - (\alpha, \gamma')\delta'$, and define $c_k = T'^k c_0 T'^{-k}$ for $k \geq 0$.

Theorem 4.5 (Explicit affine family). *Every c_k is a full-support terminal element of $W(\tilde{E}_8)$ that is not fully commutative, with*

$$L(c_k) = R(c_k) = I', \quad \ell(c_k) = 27 + 92k.$$

For every fully commutative $x \in W(\tilde{E}_8)$ and every $k \geq 0$, $\mu(x, c_k) = 0$.

Proof. The vector δ' is the null root and γ' is a root of the finite E_8 parabolic subsystem obtained by omitting node 7. Multiplying out the displayed word gives $c_0^2 = 1$. The columns $c_0(\alpha_j)$ of the matrix of c_0 recorded in the affine certificate (Appendix A; the certificate stores the matrix by rows) are negative precisely at the indices in I' , and for every edge incident to I' the sum of the two columns is a positive root; by (2) and involutivity, c_0 is terminal on both sides. Put $d = (2, -1, 1, -1, 1, -1, 1, -1, -1)$. Then $(\alpha_j, \gamma' - c_0\gamma') = d_j$, and conjugation gives

$$c_k(\alpha_j) = c_0(\alpha_j) + k d_j \delta'. \quad (5)$$

As d is negative on I' and positive off I' , like the base columns, both descent sets stay I' ; on every edge $s \sim t$ with $s \in I'$ one has $d_s + d_t \geq 0$, so (5) preserves positivity of $c_k(\alpha_s + \alpha_t)$, and c_k is terminal. Full support holds at $k = 0$ by the displayed word and at $k \geq 1$ because each descent column is strictly negative in every coordinate, whereas omitting a generator would leave its row unchanged. The c_k are pairwise distinct by (5), since every $d_j \neq 0$.

For the length, count affine inversions: for each of the 240 finite roots η write $c_0\eta = \eta' + h\delta'$, let $\epsilon(\eta)$ be 0 for positive and 1 for negative η ; the positive affine roots $\eta + m\delta'$, $m \geq \epsilon(\eta)$, contribute $\max(0, A_\eta + k B_\eta)$ to $\ell(c_k)$, where $A_\eta = \epsilon(\eta') - h - \epsilon(\eta)$ and $B_\eta = -\sum_j \eta_j d_j$. The nonzero contributions, grouped by (A_η, B_η) , are

(A, B)	$(0, 1)$	$(0, 2)$	$(1, 0)$	$(1, 1)$	$(1, 2)$
Multiplicity	44	8	1	20	6

and the remaining 161 vanish for all $k \geq 0$; the sum is $27 + 92k$.

By Remark 2.6, I' is a maximum independent set, of size 5, and $\ell(c_k)$ is odd, so Proposition 2.4 gives $\mu(x, c_k) = 0$ for every fully commutative non-cover x . Let x be a fully commutative cover of c_k . If $I' \subseteq L(x) \cap R(x)$, then Lemma 2.3 gives $x = i(I')$ of length $5 < \ell(c_k) - 1$. Otherwise, by the lifting property [3, Proposition 2.2.7], $x = s c_k$ or $x = c_k s$ for some $s \in I'$, as in the proof of Lemma 2.8. For $k \geq 1$ these elements have length $26 + 92k \geq 118$, exceeding the maximum length 44 of a fully commutative element of $\tilde{E}_8$ (complete catalogue of 44,199 elements, Appendix A, in agreement with [2, §5.2]). For $k = 0$, every cover arises by deleting one letter from the displayed reduced word, by the strong exchange condition [3, Theorem 1.4.3]; the deletions give exactly 21 distinct elements of length 26, none fully commutative (Appendix A). Hence $\mu(x, c_k) = 0$ for every fully commutative x . Finally, $c_k \notin W_{\text{FC}}$, since otherwise Lemma 2.3 would force $c_k = i(I')$. (Lemma 2.8 does not apply directly: c_0 is not a reflection.) $\square$

4.3 The E_{10} family

The first indefinite rank E_{10} lies outside the sequence E_{4r+1} . Embed the affine subdiagram on $\{0, \dots, 7, 9\}$ and take $\hat{\delta} = (2, 4, 6, 5, 4, 3, 2, 1, 0, 3)$, $\hat{\gamma} = (1, 2, 3, 2, 2, 1, 1, 0, 0, 1)$ (so $\hat{\gamma}$ differs from γ' of Section 4.2), $\hat{\beta}_0 = (3, 7, 10, 9, 7, 6, 4, 3, 1, 6)$, $\hat{I} = \{1, 3, 5, 7, 9\}$, $\hat{T} = r_{\hat{\gamma}} r_{\hat{\gamma} + \hat{\delta}}$, and for $k \geq 0$

$$\hat{\beta}_k = \hat{\beta}_0 - k\hat{\gamma} + (k^2 + 3k)\hat{\delta}, \quad \hat{b}_k = r_{\hat{\beta}_k}. \quad (6)$$

Proposition 4.6. *The reflections $\hat{b}_k$, $k \geq 0$, are pairwise distinct full-support terminal elements of $W(E_{10})$ that are not fully commutative, with $L(\hat{b}_k) = R(\hat{b}_k) = \hat{I}$. For every fully commutative x and every $k \geq 0$, $\mu(x, \hat{b}_k) = 0$.*

Proof. A lowering word in the certificate (Appendix A) reduces $\hat{\beta}_0$ to α_9 ; $\hat{\gamma}$, $\hat{\gamma} + \hat{\delta}$ are real roots of the embedded affine subsystem; and $(\hat{\beta}_0, \hat{\gamma}) = -3$, $(\hat{\beta}_0, \hat{\delta}) = -1$, so Lemma 4.2 gives $\hat{\beta}_k = \hat{T}^k \hat{\beta}_0$ with nonnegative increments $4\hat{\delta} - \hat{\gamma}$, $2\hat{\delta}$, whence $\hat{\beta}_k > 0$ in every coordinate. The pairings

$$((\alpha_j, \hat{\beta}_k))_{j=0}^9 = (-1, 1, -2-k, 1+k, -1-k, 1+k, -1-k, 1+k, -1-k^2-3k, 2+k)$$

are positive exactly on $\hat{I}$ with nonpositive sums over adjacent entries, giving both descent sets and terminality by (2); $(\alpha_1, \hat{\beta}_k) = 1$ gives full support (column 1 changes every row) and $(\hat{\beta}_k)_0 = 2k^2 + 5k + 3$ distinctness. $\hat{I}$ is a maximum independent set of odd size 5 (Remark 2.6) and $\hat{\beta}_k$ has ten nonzero coordinates, so Lemma 2.8 gives $\mu(x, \hat{b}_k) = 0$ for every fully commutative x , and $\hat{b}_k \notin W_{\text{FC}}$ by Lemma 2.7(iii). $\square$

Standard parabolic embeddings transport the $\tilde{E}_8$ and E_{10} families to larger E_n , with the supports of the smaller parabolic subgroups.

5 Complements and open problems

5.1 Necessity of the fully commutative hypothesis

A complete table of Kazhdan–Lusztig polynomials of $W(E_6)$ (51,840 elements; Appendix A) gives $\max \mu(x, w) = 10$ over all pairs, attained for instance by $x = 01254012010$ ($\ell = 11$, not FC) and $w = 123012523412301251234012$ ($\ell = 24$, full support), with

$$P_{x,w} = 1 + 13q + 59q^2 + 121q^3 + 125q^4 + 61q^5 + 10q^6.$$

Exactly 1,386 pairs have $\mu \geq 2$ (400, 556, 310, 108, 4 and 8 pairs with $\mu = 2, 3, 4, 5, 6, 10$), all with non-FC lower endpoint, while all 6,431 nonzero values with fully commutative lower endpoint equal 1, a brute-force confirmation of Theorem 1.1 for E_6 (by parabolic inclusion, $\mu \geq 10$ occurs in E_7 and E_8 as well). Full commutativity of the upper endpoint alone does not suffice either: $x = 5343010$ (not FC) and $w = 0125342312501234$ (FC, $\ell = 16$) have

$$P_{x,w} = 1 + 8q + 22q^2 + 20q^3 + 5q^4, \quad \mu(x, w) = 5.$$

5.2 The bound fails in affine type

In $\tilde{D}_4$, with x the product of the four leaves and c the centre, $w = xcx$ has $P_{x,w} = 1 + 3q + 2q^2$ and $\mu(x, w) = 2$; this star family is covered by Mongelli [29, Corollary 2 (arXiv v1, p. 10)], and the value is recomputed in Appendix A. Hence Corollary 1.2 is sharp within simply laced type, and the universal form of Green’s question has a negative answer. Vanishing of μ does not mean $P = 1$: in finite D_4 , with x the product of the three leaves, $b = xcx$ has maximum common commuting descents and even gap 4, so $\mu(x, b) = 0$, but $P_{x,b} = 1 + 2q$ [29, Corollary 3 (arXiv v1, §4)].

5.3 Terminals in affine E_8 beyond the family

The reflection

$$w = r_\beta = 875634523412301282341230128567453, \quad \beta = (1, 2, 3, 3, 2, 2, 1, 1, 2),$$

of length 33 is a full-support terminal with $L(w) = R(w) = \{3, 5, 7, 8\}$, which is not a maximum independent set, so Proposition 2.4 does not apply. It has exactly five eligible fully commutative lower endpoints, all with $\deg P_{x,w} = 13$:

x	8753	87530	87531	875301	875310
gap $\ell(w) - \ell(x)$	29	28	28	27	27
$\mu(x, w)$	0	0	0	1	1

The right extensions ws_0 and ws_1 of length 34 are terminal involutions with $L = R = \{0, 3, 5, 7, 8\}$, respectively $\{1, 3, 5, 7, 8\}$, each a maximum independent set, each with a unique eligible fully commutative lower endpoint (87530, respectively 87531, gap 29, $\deg P = 14$) and $\mu = 1$, so these two pairs satisfy every hypothesis of Proposition 2.4 except parity. All values found lie in $\{0, 1\}$. The polynomials and the sizes of the lower ideals and quotients are recorded in the certificate (Appendix A); the computation runs the Kazhdan–Lusztig recursion on the parabolic quotient $W/W_{R(w)}$, which is legitimate because $P_{x,w} = P_{xs,w}$ for $s \in R(w)$ [25, §2.3].

5.4 Conditional interval transfer

Combinatorial invariance, conjectured independently by Lusztig and Dyer [9], asserts that $P_{x,w}$ depends only on the poset $[x, w]$; it is known for lower intervals [5, 6, 8], for the coefficient of q [31, 1] and for short intervals [23, 1, 11] (see also [4]), and a complete proof for arbitrary Coxeter systems is claimed in the unrefereed machine-generated OpenAI preprint [30, Theorem 1.1]. Assuming it, the bound of Corollary 1.2 and the vanishing for the families of Section 4 hold for every Bruhat interval in any Coxeter group that is isomorphic as a poset to one of the intervals concerned; the formal conditional consequences for Gern’s intervals $[x_n, w_n]$ in D_n are Theorem 2.1 and Corollary 2.2 of the archived note `kl-transfer` (Appendix A). No new interval isomorphisms are asserted.

5.5 Open problems

(1) Is $\mu(x, w) \in \{0, 1\}$ for every fully commutative x and every $w \in W(\tilde{E}_8)$? All values computed so far, including those of Section 5.3, lie in $\{0, 1\}$. (2) Classify the terminal elements of $\tilde{E}_8$, and of the indefinite groups E_n , up to translation. (3) Determine μ for the translated length-33 family $r_{\beta+k\delta'}$, $k \geq 1$. (4) Settle Green’s question for F_4 , H_3 and H_4 . (5) Give an effective description of all noncommuting terminals of E_n for $n \geq 9$; Theorem 1.3 shows that no finite list exists.

Use of AI tools

This work was carried out with substantial assistance from large language model systems. OpenAI’s GPT-6.1 Sol and GPT-6 Astra, run as teams of agents, performed the literature searches, proposed and drafted the constructions and proofs, wrote the enumeration and verification code, produced internal review notes, and drafted the text. Anthropic’s Claude Fable 5.1 then carried out an independent adversarial review: it re-derived every lemma against the primary sources, recomputed every computational claim with separately written code, checked every citation, and found the strengthening of Theorem 1.3, the data in Section 5, and the attribution corrections incorporated here; the Go port of the supplement was written by Claude Sonnet 5.5 and verified by Claude Fable 5.1 against the earlier programs’ outputs. The author directed the work, checked the proofs, computations and references, and takes full responsibility for the content. The complete workflow, including the prompts used, the internal review notes (which are machine-generated and are not peer review), and the review code, is recorded in the repository cited in Appendix A.

A Computational certificates

The following computations are parts of proofs: the terminal classification of Proposition 3.1; the reducedness, involution and descent data of the words in Table 1 and of c_0 , whose matrix (hence the columns $c_0(\alpha_j)$) is recorded; the D_6 polynomial $P_{x_6, w_6} = 1 + 6q + 11q^2 + 6q^3 + q^4 + q^5$; the complete fully commutative catalogue of $\tilde{E}_8$ (44,199 elements, maximum length 44) and the cover check for c_0 ; and the lowering words certifying the real roots γ , $\gamma + \delta$, $\hat{\beta}_0$, $\hat{\gamma}$ and $\hat{\gamma} + \hat{\delta}$. Everything else is a cross-check or a complement: the fully commutative catalogues of E_6 , E_7 , E_8 (agreeing with [34] and [2, §5.1]), Poincaré-polynomial

Result	Programs; certificates
E_6, E_7	enumerate-bad, matrix-search, verify-exceptional; e6_bad.json, e7_bad.json, e6-independent-certificate.json, e7-independent-certificate.json
E_8	terminals-flat, terminals-recursive (both on internal/parabolic), verify-outputs, e8-d7, verify-e8; e8-recursive.json [†] , e8-d7-terminals.json [†]
Table 1, Remark 3.3	finite-descents, terminal-structure*; finite-descents.json, terminal-structure-certificate.json
D_6, D_8	d6-certificate (on internal/d6), d8-germ-kl*; d6-recurrence-certificate.json, d8-germ-certificate.json
Uniform family	uniform-verify, uniform-covers*, fc-maxima*; referee-certificate.json [†] , uniform-family-certificate.json, fc-maxima-certificate.json
Affine $\tilde{E}_8$	affine-proof, affine-fc-covers (both on internal/affine), fc-catalogue; proved-affine-certificate.json, fc-cover-certificate.json [†]
E_{10}	e10-all-k (on internal/indefinite); e10-certificate.json [†]
Section 5	e6-mu-table*, e9-quotient-kl*, affine-d4; e6-mu-table.json, e9-odd-gap-certificate.json, affine-d4-independent.json

Table 2: Programs (the commands `cmd/name` of the Go module) and certificates of the supplement (the zip of release v0.3.0), by result. Starred programs were written for the review and share no code with the others; the `internal` libraries in parentheses are the only shared code; daggered certificates are written by the runner, not shipped as inputs.

checks, finite-rank checks of the uniform family, the structure data of Remark 3.3, the fully commutative catalogues and maxima of E_{10} – E_{13} , the data of Section 5, and the D_8 polynomial $P_{x_8, w_8} = P_{e, w_8} = 1 + 12q + 59q^2 + 154q^3 + 233q^4 + 221q^5 + 147q^6 + 70q^7 + 20q^8 + 2q^9$ of the archived note `kl-transfer`. All arithmetic is exact. For E_6 and E_7 two programs sharing no code (root indices versus integer columns) give full enumerations agreeing with the Poincaré products; for E_8 the recursive pruning engine and a separate enumeration along a D_7 parabolic reproduce the same list, sharing only their packed root representation, an integer-matrix verifier rechecks every terminal word, length, terminal test and eligible lower endpoint, and the reviewers’ own enumeration along four parabolic chains, among them the E_7 and D_7 chains above (code under `research/claude-review-2026-10-07/verification-code/` in the repository, not part of the zip), gives the same list. The D_6 polynomial comes from a sparse Kazhdan–Lusztig recursion checked through all 24,245 dependency records of the saved certificate (a consistency check, the checker sharing group arithmetic with the generator); `d8-germ-kl` recomputes it with independent code in the signed-permutation model and, in the extended run, in the geometric representation as well, and the reviewers’ code just cited recomputes it in both models and directly inside E_7 and E_8 . The uniform root proof is symbolic; a verifier checks Lemmas 4.1–4.3 symbolically and the finite-rank matrices for $3 \leq r \leq 40$. The starred programs of Table 2, written for the review, share no code with the others.

Programs, inputs and output certificates (Table 2) are archived with SHA-256 checksums in release v0.3.0 of the repository `github.com/tygern/kl` (zip `exceptional-leading-proof.zip`); a single command rebuilds every classification and verifies every certificate with Go 1.22 or later and no other dependency. The earlier Python and C++ programs remain in release v0.2.0 and under `research/archive/python-cpp/`, and the Go port was verified against their outputs certificate for certificate. Copies of the daggered certificates of Table 2, which the runner writes rather than reads, are kept in the repository. The archived research notes of Section 5 are in `research/archive/notes/`.

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